

A prospective assay for hypothesis-space expansion in AI systems

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Abstract

Scientific discovery can require changing not only a hypothesis but the language in which hypotheses are expressed. Existing evaluations rarely make that boundary, or the cost of crossing it, objectively testable. We introduce a prospective assay for bounded hypothesis-space expansion. A candidate must be structurally outside a frozen incumbent language, fit existing observations, commit to a discriminating intervention before its outcome is revealed, outperform the best incumbent predictor and survive independent falsification. Across synthetic mechanistic worlds, typed high-level proposals succeeded in 142 of 400 cases, whereas direct, sampled, fixed-space and attribute-only alternatives succeeded in 0-1. Compositions of generic local rewrites succeeded in 400 of 400 known-family and 100 of 100 held-out-family worlds. A post-hoc component audit reproduced all C3 verdicts without language-model outputs, locating this result in the deterministic search scaffold rather than the model. The assay enables auditable attribution of hypothesis-space expansion while exposing saturation and limited external validity.

Introduction

Abduction selects an explanatory hypothesis, but scientific change sometimes alters the representational vocabulary in which hypotheses can be stated¹⁻³. Computational creativity and constructive-induction research formalized related distinctions between exploration within a space and transformations of that space⁴⁻⁹. Symbolic regression, program search and language-model-guided systems can discover useful equations, algorithms and hypotheses¹⁰⁻¹⁷, while multi-agent systems increasingly automate broader scientific workflows¹⁸⁻²⁰.

Recent work now explicitly targets expanding principle or model spaces. PiEvo treats discovery as optimization over an evolving principle space across four benchmarks²¹. Model Discovery Agent couples a language-model proposer to Bayesian experiment design in an open model setting spanning physics, chemistry and biology²². HypoArena evaluates prospective hypothesis discovery in 988 cases across six domains and 15 frontier models²³. These studies broaden the scientific tasks and models under evaluation. A complementary measurement problem remains: how can an evaluation prove that a candidate left a specified incumbent hypothesis language, charge it a prediction before seeing the answer, and separately identify which system component caused the escape?

Here we operationalize bounded hypothesis-space expansion. We freeze an incumbent representational language, generate executable candidates and require a six-gate verdict: incumbent adequacy (J0), canonical structural non-membership (J1), candidate adequacy (J2), a discriminating committed prediction (J3), prospective intervention gain (J4) and independent falsification survival (J5; Fig. 1). Candidate prose, model confidence, embedding distance and semantic novelty do not enter the verdict.

Our experiments validate the assay and reveal a component-attribution limit. Typed proposals and generic compositional search cross the frozen language reliably, whereas fixed-space alternatives do not. However, a code-path audit and inference-free replay show that C3's representations, fitted

equations, structural ranking and interventions were deterministic; removing the model output leaves all 500 jump and 300 control verdicts unchanged. The central contribution is therefore the prospective assay and a transparent typed search scaffold-not evidence that a language model was necessary for the successful escapes.

Table 1 | Position relative to adjacent discovery evaluations

Evaluation	Exec	Frozen	Non-member	Prospective	Falsify	Attribution / replay	Breadth
Hypothesis Search / HypoGen	partial	no	no	no	no	no	language tasks
POPPER	partial	no	no	yes	yes	limited	literature
FunSearch	yes	skeleton	no	feedback	tests	proposer/evaluator	mathematics
PiEvo	yes	evolving	no	varies	varies	proposer/search	4 benchmarks
Model Discovery Agent	yes	open set	no	Bayesian	predictive	proposer/inference	3 sciences
HypoArena	judged	no	no	context	rubric	no	988 / 6 / 15
This work	yes	yes	canonical	locked	exact	factorial / exact	9 families / 1 model

Results

A prospective criterion for hypothesis-space expansion

Figure 1 | A prospective assay for bounded representation escape

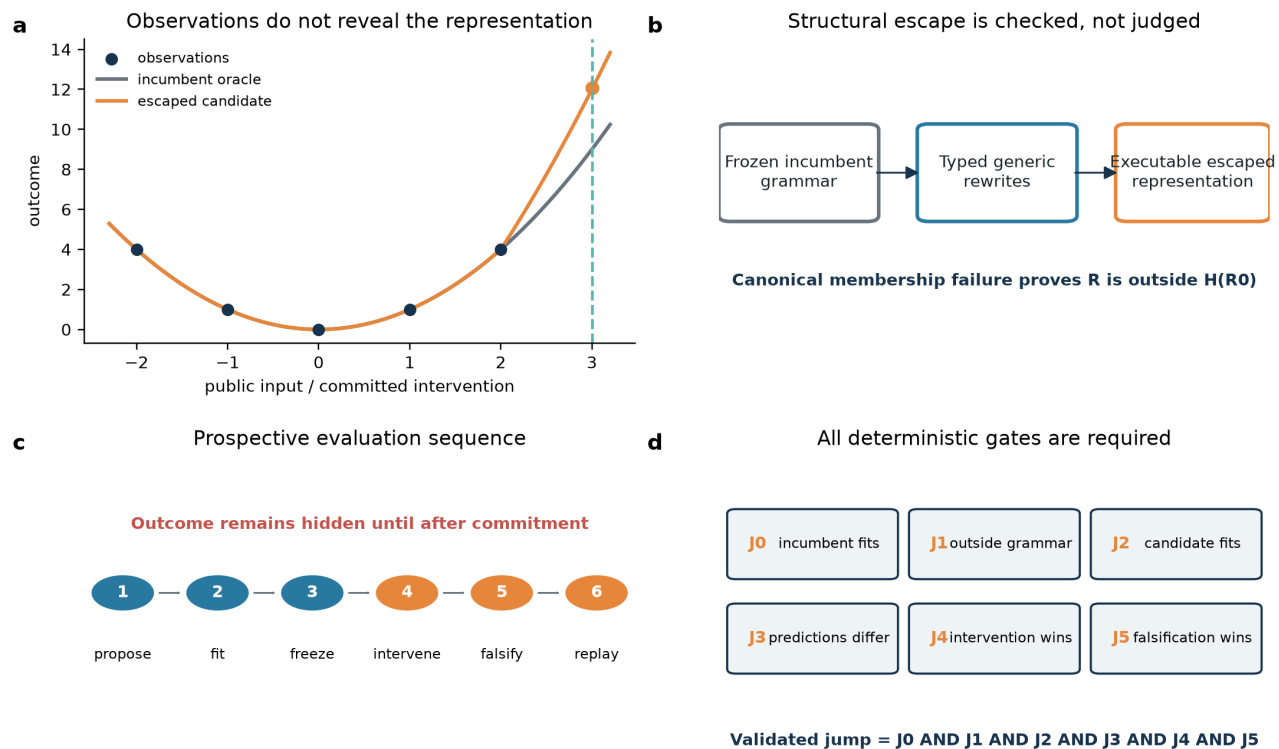


Figure 1 | The assay combines canonical structural non-membership with a prospective intervention and independent falsification. Panel a is schematic; panels b-d summarize the registered pipeline.

Each synthetic world contains observations compatible with an incumbent model, a frozen typed-graph language, hidden intervention outcomes and independent falsification cases. The best incumbent predictor is computed exactly or by bounded exhaustive search. Before any intervention outcome is

revealed, a candidate representation, executable expression, selected action, prediction and split hash are committed. A validated jump is the conjunction $J0 \wedge J1 \wedge J2 \wedge J3 \wedge J4 \wedge J5$ (Fig. 1). This design turns "leaving a hypothesis space" into a replayable event rather than a judgement about wording or apparent novelty.

Typed proposals outperform fixed-space alternatives

Figure 2 | Typed proposals and their gate attrition

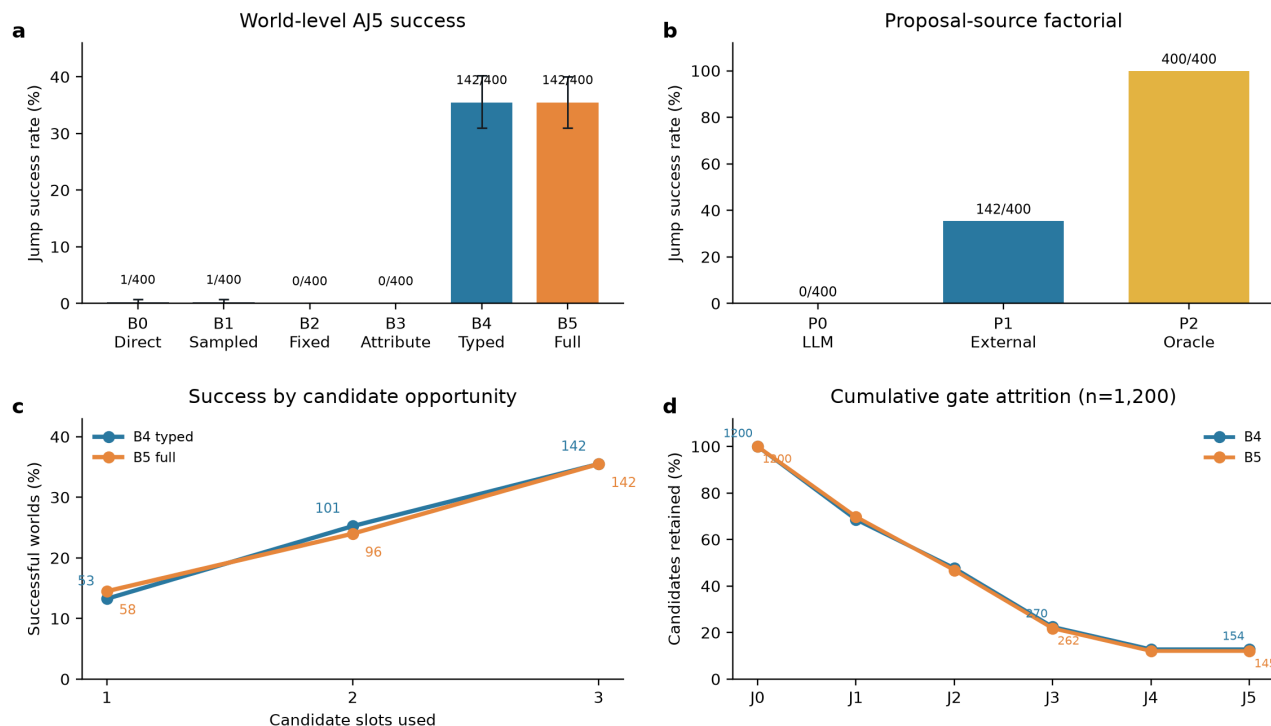


Figure 2 | AJ5 world-level jump success. Error bars are registered family-stratified bootstrap 95% intervals. Counts are successful worlds; $n=400$ per jump condition. All conditions recorded 0/200 false jumps.

In AJ5, direct model proposals (B0) and sampling-matched proposals (B1) each succeeded in 1 of 400 worlds. Fixed-space reasoning (B2), attribute-only mutation (B3) and value-only mutation succeeded in none. External typed representation proposals succeeded in 142 of 400 worlds in both B4 and B5 (35.5%; Fig. 2a). All eight registered comparisons of B4 or B5 against B0-B3 were positive after Holm correction (adjusted $P \leq 4.00 \times 10^{-4}$).

The proposal-source factorial held the downstream path fixed. Model-proposed representations (P0) succeeded in 0 of 400 worlds, external typed proposals (P1) in 142, and oracle representations (P2) in 400 (Fig. 2b). This shows that the tested system can execute a supplied changed representation. It does not establish that the model is needed after deterministic fitting, because the fitted expression and maximum-separation intervention were supplied to the second call and programmatically enforced.

Candidate-level gate attrition clarifies the result (Fig. 2c). In jump worlds, B4 had 1,200 candidates at J0, 823 through J1, 573 through J2, 270 through J3 and 154 through J4-J5. B5 had 1,200, 838, 562, 262 and 145, respectively. In controls, B4 and B5 still produced 112 and 110 candidates through J3, but none passed J4. The zero-control result is therefore not created by J0-J3 alone; prospective outcomes eliminate apparently admissible escapes.

Generic rewrites compose into validated representations

Figure 3 | Generic rewrites compose into validated representations

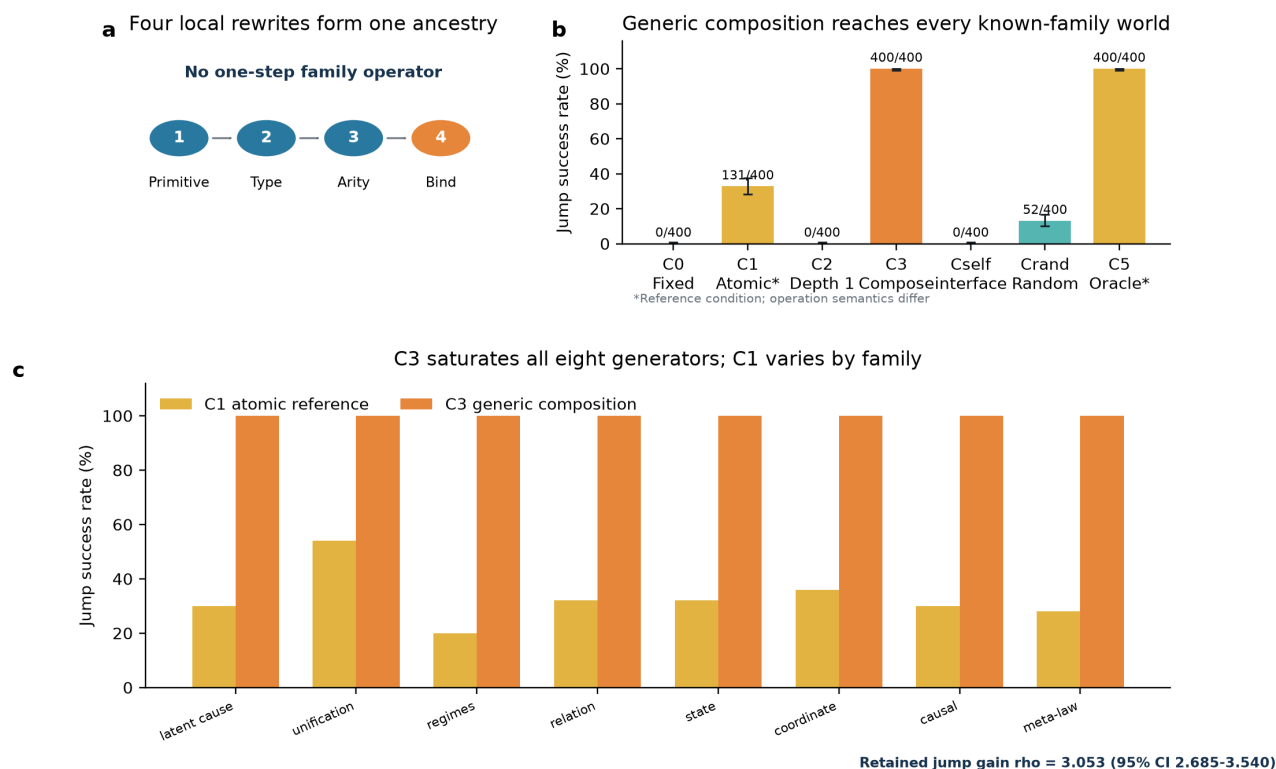


Figure 3 | CJ5 known-family reconstruction. Error bars in panel b are Wilson 95% intervals; $n=400$ worlds per condition. C1 and C5 are reference conditions with different operation semantics. Per-family panels contain 50 worlds each.

CJ5 replaced nine high-level mutations with 29 local graph and abstract-syntax-tree rewrites. Adding a node, changing its type or observability, changing arity and binding each argument were separate steps. No primitive accepted a family label, target distance or outcome. C3 deterministically traversed 48 four-step branches, ranked candidates using observational fit, structural escape and prospective prediction separation without outcomes, and retained three.

Across eight known structural families with new seeds ($n=400$ worlds), fixed-space search (C0) and 192 generic depth-one alternatives (C2) succeeded in no worlds. The frozen atomic reference (C1) succeeded in 131 (32.75%), random four-step paths (C_rand) in 52 (13.0%), and C3 in all 400 (Fig. 3). C3 exceeded C_rand by 0.87 (95% CI 0.845-0.895; Holm-adjusted $P=3.00 \times 10^{-4}$). C3 was 100% in every family; this demonstrates within-generator reliability and assay separability, not eight statistically independent replications or an unsaturated frontier benchmark.

C_self also scored 0/400, but its failure is narrower than initially anticipated. All 38,400 confirmatory plan records failed the same schema check because plans was not a list; no proposed four-step plan reached structural evaluation. C_self therefore measures the registered serialization interface, not conceptual inability to compose representations. Grammar-constrained decoding and validator repair remain necessary future controls.

A deterministic component audit removes the language model

Figure 4 | Transfer, depth and model-free replay

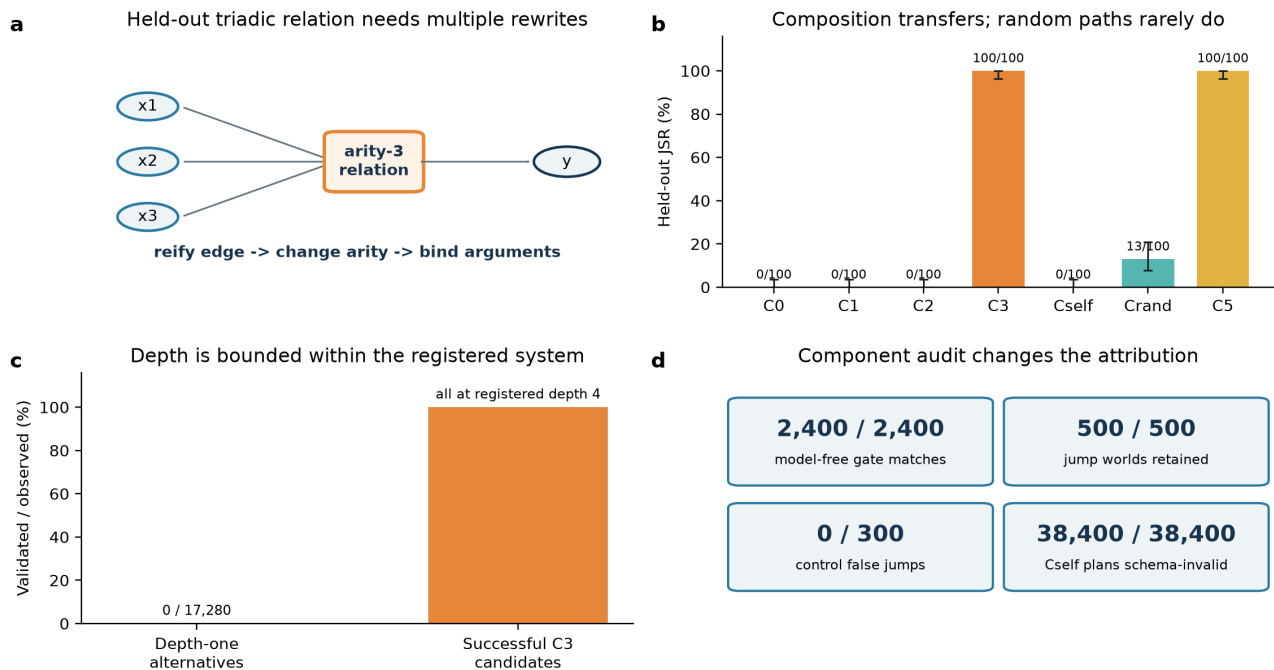


Figure 4 | Held-out transfer, depth bounds and the post-hoc deterministic component audit. The model-free replay reproduced all 2,400 C3 candidate verdicts and retained 500/500 jump successes with 0/300 control false jumps. Cself never reached structural evaluation because every plan record failed the same schema check.

Inspection of the frozen C3 execution path showed that the representation, expression, candidate ranking and selected intervention were generated deterministically before the model response. The response's representation, expression and intervention fields were overwritten; only its explanation remained, and explanation does not enter J0-J5. We therefore performed a post-hoc, inference-free ablation that replaced both model calls with a valid empty explanation and recomputed every C3 theory and commitment from the archived representation, expression and intervention.

All 2,400 candidate verdicts matched the archive. The model-free replay retained 500/500 jump-world successes and 0/300 control-world false jumps (Fig. 4). Thus Phi-4 was not causally necessary for C3 under the implemented path. AJ5 still measures differences among model-facing proposal interfaces, but the perfect C3 result belongs to the typed deterministic search-and-evaluation scaffold. This audit changes the system attribution, not the validity of the prospective assay.

Transfer to one held-out structural family

The second study reserved triadic_relation_reification from AJ5 development, CJ5 pilot inference and known-family confirmation. Observations supported an incumbent cubic rule under correlated inputs; the target reified an arity-three relation exposed by intervention. C3 succeeded in 100 of 100 held-out worlds, C_rand in 13, and C0, C1, C2 and C_self in none. C3 exceeded C_rand by 0.87 (95% CI 0.80-0.93; adjusted $P=3.00 \times 10^{-4}$).

The holdout is limited. The earlier study contained a binary property-to-relation family, and the generic language already contained reification. What was new was the arity-three structure, product mechanism, generator and instances. Success across a single conceptually adjacent family cannot support broad external generalization.

A worked prospective escape

Figure 5 | One complete prospective escape

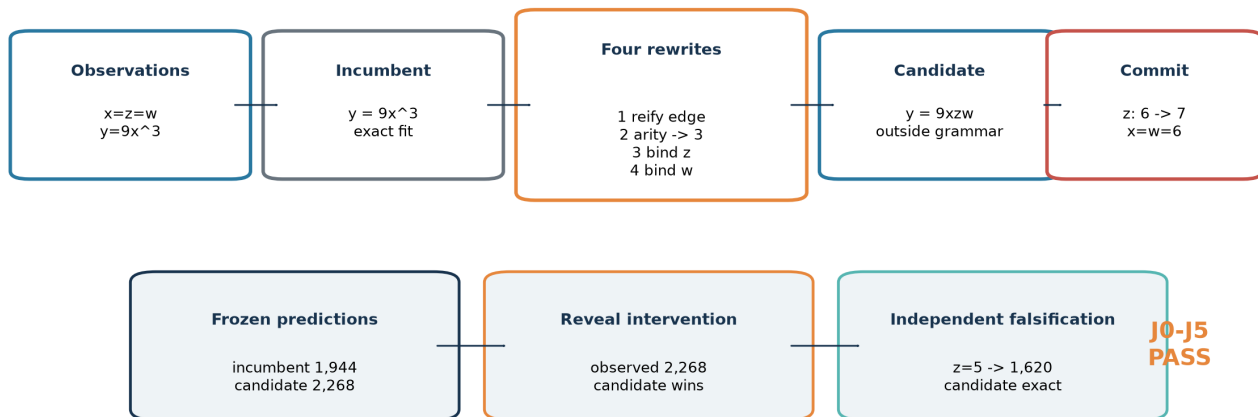


Figure 5 | Worked held-out example. Correlated observations make the cubic incumbent and triadic candidate observationally identical. A committed intervention separates them before outcome reveal, and an independent case falsifies the incumbent.

In one held-out world, observations with $x=z=w$ gave $y=9x^3$, so the incumbent $y=9x^3$ and candidate $y=9xzw$ both fit exactly. Four generic edits reified the incumbent prediction edge, changed the new relation to arity three, and bound z and w (Fig. 5). Before outcome reveal, the assay committed to setting z from 6 to 7 at $x=w=6$. The incumbent predicted 1,944; the candidate predicted 2,268; the revealed outcome was 2,268. Independent falsification at $z=5$ yielded 1,620, again exact for the candidate. The representation failed the frozen language on relation count, type, arity bindings and result structure, and passed J0-J5.

Specificity and replay

The zero false-jump count requires qualification. In C3 controls, 800 of 900 candidates passed J1, 567 passed J2 and 283 passed J3, but none passed J4. The null was therefore non-trivial through the prospective commitment gate. Nevertheless, because no-jump truth is generated inside the incumbent language and the incumbent oracle is exact, an outside candidate cannot strictly beat it under noiseless evaluation. The observed 0/300 is best interpreted as a structural-specificity check under the exact simulator, not an empirical false-positive estimate for noisy science; the Wilson upper bound is descriptive only.

All 10,800 AJ5 and 16,800 CJ5 selected candidates replayed exactly, including 35,533 CJ5 ancestry records. No confirmatory world was excluded and no shard was rerun.

Discussion

This work contributes an assay for a specific event: an executable candidate leaves a frozen hypothesis language and pays for that move with a prospective prediction that beats the best incumbent and survives independent falsification. Canonical non-membership, outcome-before-commitment ordering, proposal-source factorials, false-jump controls and deterministic replay distinguish the assay from broader discovery benchmarks (Table 1).

The experiments also show why component attribution must be audited at code-path level. AJ5 indicates that typed external proposals reach useful representations more often than the registered model interfaces. But C_self never crossed its parser, and C3 did not require the model output. The

defensible mechanistic result is therefore that typed representation search can cross a frozen space and that interface failures can be separated from downstream executability. Claims of language-model-driven representation change would require a redesigned path in which model-generated scientific content survives into the committed theory, plus a true minus-model ablation.

External validity is the principal limitation. The worlds are synthetic and noiseless; C3 is saturated; search strata and generators were co-designed; the operator language embeds graph, function, relation and binding priors; only one model and one adjacent held-out family were tested. World seeds quantify within-generator reliability. The higher-level generalization units are nine structural families, only one of which was held out, so small world-level P values do not substitute for structural breadth.

A decisive extension should freeze the current assay and add independently authored families, distractor primitives, deeper compositions, stochastic observations, partial observability and at least one recognizable mechanistic simulation. It should compare a stronger open model and a frontier model, with raw, grammar-constrained and validator-repair interfaces, and report parse validity and cumulative J0-J5 attrition. These experiments are proposed, not part of the present evidence.

The current conclusion is narrower but useful: hypothesis-space expansion can be measured prospectively and replayed exactly; typed deterministic search can produce it in controlled worlds; and the assay exposes when an apparent language-model capability is actually supplied by the surrounding scaffold.

Methods

Study design and protocol freezing

AJ5 and CJ5 protocols were frozen in Git before their reported confirmatory model calls. AJ5 uses commit 895ebb9118ffd0046825b88868621f2a70f69f61. CJ5 uses 65f20874e16bddf8a7ae36996395ff52b27153b7, with a documented correction at 7ecb977 and held-out unlock at 27ee542. The commits are publicly retrievable but unsigned, and no independent registry or transparency-log timestamp was located. We therefore describe the studies as prospectively specified and commit-frozen, not formally preregistered. Pilot seeds were excluded; no confirmatory result changed families, operators, thresholds, budgets or seeds.

Model and inference

Both studies used frozen microsoft/phi-4 revision 2db69c1c3e91a05d2c64a3185acfbaf36f744e25, vLLM 0.10.2, dynamic bitsandbytes 4-bit quantization, a 4,096-token context, temperature 0.2, top-p 0.95 and completion cap 700. AJ5 sampling proposals used temperature 0.7. Each request had a deterministic seed. No fine-tuning or cross-world adaptation occurred.

Worlds, representations and search

Generators returned public observations, redacted actions, a frozen incumbent LanguageSpec, hidden truth and independent intervention and falsification cases. AJ5 used eight families and 50 jump seeds per family; CJ5 used 50 new seeds per family. No-jump controls used 25 seeds per family in each phase. CJ5 added 100 held-out jump and 100 control seeds. Representations were canonical typed graphs with executable fitted programs.

AJ5 tested direct model output, model sampling, fixed-space reasoning, attribute-only mutation and two typed high-level proposal conditions. CJ5 used 29 generic local rewrites. C3 traversed 48 four-operation branches and retained three candidates by a deterministic score. C_rand sampled 48 four-step paths. C_self requested 16 plans in each of three slots; invalid plans were not repaired.

Prospective evaluation

J0 required incumbent observation $MSE \leq 10^{-12}$; J1 required DSL validity and frozen-language non-membership; J2 required candidate observation $MSE \leq 10^{-12}$; J3 required candidate-oracle prediction separation ≥ 0.5 on the committed action; J4 required intervention MSE improvement > 0.1 ; J5 required falsification $MSE \leq 10^{-12}$ and improvement > 0.1 . Candidate, prediction, action and split hashes were committed before simulator evaluation.

Statistics and component audit

The world was the replicate. AJ5 used 10,000 family-stratified paired bootstrap replicates; CJ5 used the same bootstrap for effects and paired sign-flip tests. Holm correction was applied within prospectively specified comparison families at $\alpha = 0.05$. Candidate rows were used only for attrition. Family-level distributions are reported descriptively because eight known families plus one held-out family do not support a stable population-level family inference.

The post-hoc component audit reconstructed all C3 candidates from archived deterministic representations, fitted expressions and selected interventions, replacing the model-derived explanation with an empty string. It then regenerated each world, froze a new commitment and recomputed J0-J5. No model inference was run.

Integrity and replay

Requests, representations, ancestry edges, fitted programs, commitments, gate values and configuration hashes were retained. Replay reconstructed representations and recomputed J0-J5. Infrastructure failures would have required whole-shard reruns; none occurred. There were no outcome-quality exclusions.

AI assistance in research and writing

Phi-4 generated the registered outputs described above. OpenAI Codex was used only after experiments to inspect artifacts, recompute inference-free audits, verify literature metadata and assist drafting. It did not choose confirmatory hypotheses, alter source data or run new model inference. Human authors verified the cited metadata and remain responsible for originality, accuracy and integrity.

Data availability

Synthetic-world definitions, result tables, manifests and replay artifacts are included in the project repository. The submission release will archive an exact commit in a DOI-minting repository. No personal or restricted third-party data were used.

Code availability

Source code for generation, search, evaluation, statistics, component audit and replay is included in the project repository. The release will identify the exact commit and archival DOI. Model weights are not redistributed; identifier, revision and runtime are reported above.

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